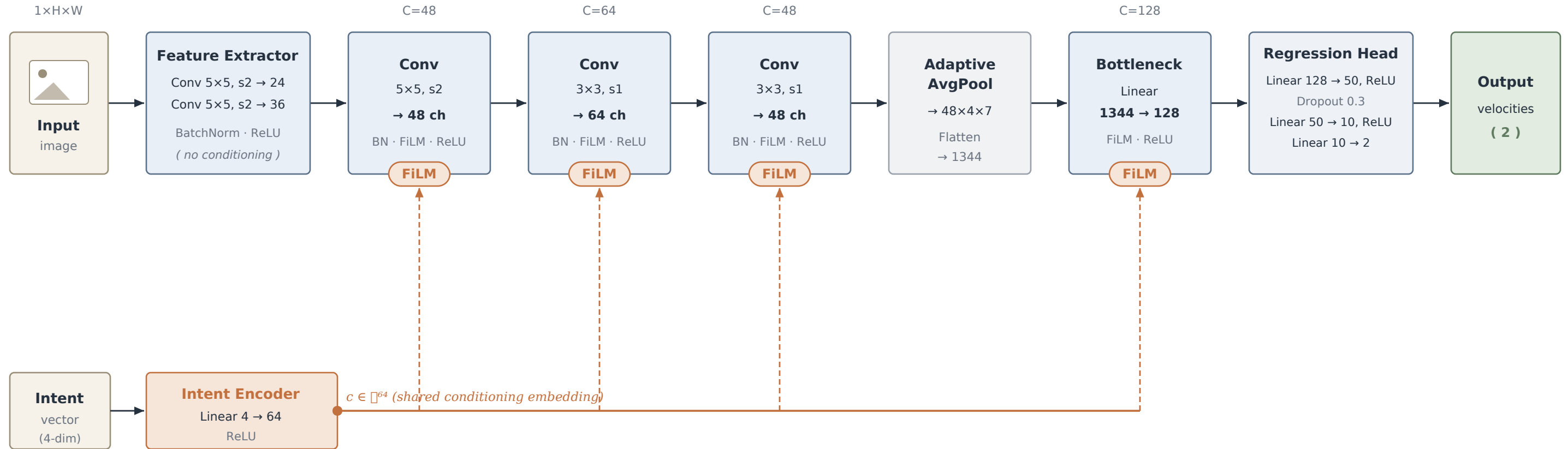


Conditional PilotNet with FiLM Intent Conditioning

grayscale image $\rightarrow$ steering / velocity, modulated by a shared intent embedding



Legend

- Image processing
- Pool / flatten
- Conditioning / FiLM
- Output

FiLM layer (per conditioned stage)

$\gamma = \text{Linear}(64 \rightarrow C)$, $\beta = \text{Linear}(64 \rightarrow C)$

$$\text{out} = \gamma \odot x + \beta$$

identity init ($\gamma=1$, $\beta=0$) $\rightarrow$ starts as a no-op;

broadcasts over H, W for conv maps; per-channel for the bottleneck.